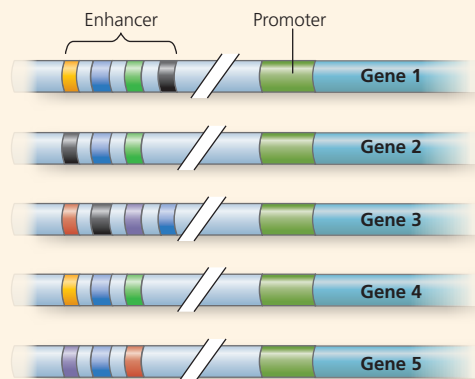


Levels 3-4: Applying/Analyzing

6. What would occur if the repressor of an inducible operon were mutated so it could not bind the operator?
(A) irreversible binding of the repressor to the promoter
(B) reduced transcription of the operon's genes
(C) buildup of a substrate for the pathway controlled by the operon
(D) continuous transcription of the operon's genes
7. Absence of *bicoid* mRNA from a *Drosophila* egg leads to the absence of anterior larval body parts and mirror-image duplication of posterior parts. This is evidence that the product of the *bicoid* gene
(A) normally leads to formation of head structures.
(B) normally leads to formation of tail structures.
(C) is transcribed in the early embryo.
(D) is a protein present in all head structures.
8. Which of the following statements about the DNA in one of your brain cells is true?
(A) Most of the DNA codes for protein.
(B) The majority of genes are likely to be transcribed.
(C) It is the same as the DNA in one of your liver cells.
(D) Each gene lies immediately adjacent to an enhancer.
9. Within a cell, the amount of protein made using a given mRNA molecule in that cell depends partly on
(A) the degree of DNA methylation.
(B) the rate at which the mRNA is degraded.
(C) the number of introns present in the mRNA.
(D) the types of ribosomes present in the cytoplasm.
10. Proto-oncogenes can change into oncogenes that cause cancer. Which of the following best explains the presence of these potential time bombs in eukaryotic cells?
(A) Proto-oncogenes first arose from viral infections.
(B) Proto-oncogenes are mutant versions of normal genes.
(C) Proto-oncogenes are genetic "junk."
(D) Proto-oncogenes normally help regulate cell division.

Levels 5-6: Evaluating/Creating

11. **DRAW IT** The diagram below shows five genes, including their enhancers, from the genome of a certain species. Imagine that yellow, blue, green, black, red, and purple activator proteins exist that can bind to the appropriately color-coded control elements in the enhancers of these genes.



- (a) Draw an X above enhancer elements (of all the genes) that would have activators bound in a cell in which only gene 5 is transcribed. Identify which colored activators would be present.
- (b) Draw a dot above all enhancer elements that would have activators bound in a cell in which the green, blue, and yellow activators are present. Identify which gene(s) would be transcribed.

- (c) Imagine that genes 1, 2, and 4 code for nerve-specific proteins, and genes 3 and 5 are skin-specific. Identify which activators would be present in each cell type to ensure transcription of the appropriate genes.

12. **EVOLUTION CONNECTION** DNA sequences can act as "tape measures of evolution" (see Concept 5.6). Scientists analyzing the human genome sequence were surprised to find that some of the regions of the human genome that are most highly conserved (similar to comparable regions in other species) don't code for proteins. Propose a possible explanation for this observation.
13. **SCIENTIFIC INQUIRY** Prostate cells usually require testosterone and other androgens to survive. But some prostate cancer cells thrive despite treatments that eliminate androgens. One hypothesis is that estrogen, often considered a female hormone, may be activating genes normally controlled by an androgen in these cancer cells. Describe one or more experiments to test this hypothesis. (See Figure 11.9 to review the action of these steroid hormones.)
14. **WRITE ABOUT A THEME: INTERACTIONS** In a short essay (100–150 words), discuss how the processes shown in Figure 18.2 are examples of feedback mechanisms regulating biological systems in bacterial cells.
15. **SYNTHESIZE YOUR KNOWLEDGE**



The flashlight fish has an organ under its eye that emits light, which serves to startle predators and attract prey, and allows the fish to communicate with other fish. Some species can rotate the organ inside and then out, so the light appears to flash on and off. The light is actually emitted by bacteria (of the genus *Vibrio*) that live in the organ in a mutualistic relationship with the fish. (The bacteria receive nutrients from the fish.) The bacteria must multiply until they reach a certain density in the organ (a "quorum"; see Concept 11.1), at which point they all begin emitting light at the same time. There is a group of six or so genes, called *lux* genes, whose gene products are necessary for light formation. Given that these bacterial genes are regulated together, propose a hypothesis for how the genes are organized and regulated.

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom AAAS

Can cancer be stopped before it begins?

Go to "Arrested Development: When Cells Make Mistakes" at www.scienceintheclassroom.org.

→ **Instructors:** Questions can be assigned in Mastering Biology.